Experiment 4

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Subject Name: FULLStack2
Subject Code: 23CSH-382

1. **AIM:** To optimize the performance of the EcoTrack React application using memoization techniques and code splitting, and to enhance the user interface using enterprise-grade Material UI components.
2. **Objective:**
   After completing this experiment and its follow-up tasks, the student will be able to:
   1. Understand the causes of unnecessary re-renders in React applications
   2. Optimize React components using React.memo to prevent avoidable re-renders
   3. Apply useMemo to efficiently compute derived data and avoid redundant calculations
   4. Use useCallback to memoize event handler functions and improve component performance
   5. Implement lazy loading of components and routes using React.lazy and Suspense
   6. Reduce initial bundle size and improve application load performance through code splitting
3. **Implementation/Code:**
   a. **logsSlice.jsx**

```
import { createSlice, createAsyncThunk } from
"@reduxjs/toolkit";
```

```
export const fetchLogs = createAsyncThunk(
   "logs/fetchLogs",
   async () => {
       await new Promise((resolve) => setTimeout(resolve,
1000));
       return[
           { id: 1, activity: "Car Travel", carbon: 4 },
           { id: 2, activity: "Electricity Usage", carbon: 6 },
           { id: 3, activity: "Cycling", carbon: 0 },
       ]
   }
)

const logSlice = createSlice({
   name: "logs",
   initialState: {
       data: [],
       status: "idle",
       error: null,
   },
   reducers: {},
   extraReducers:(builder) => {
       builder
       .addCase(fetchLogs.pending, (state) => {
           state.status = "loading";
       } ).addCase(fetchLogs.fulfilled, (state, action) => {
           state.status = "success";
           state.data = action.payload;
       }).addCase(fetchLogs.rejected, (state, action) => {
           state.status = "failed";
           state.error = action.error.message;
       })
   }
})
export default logSlice.reducer;
```

b. Slice.jsx

```
import React from 'react'
import {configureStore, createSlice} from '@reduxjs/toolkit';
import LogsReducer from './logsSlice';
```

```
const store = configureStore({
   reducer: {
       logs: LogsReducer,
   },
});
export default store;
```

c. **LogsPage.jsx**

```
import { useDispatch, useSelector } from 'react-redux';
import { fetchLogs } from '../store/logsSlice';

const Logs = () => {
   const dispatch = useDispatch();
   const { data, status, error } = useSelector((state) =>
state.logs);

   const handleFetchLogs = () => {
       dispatch(fetchLogs());
   };

   if (status === "loading") {
       return <p>Loading your logs..........</p>;
   }

   if (status === "failed") {
       return <p>Error: {error}</p>;
   }

   const logs = data || [];

   return (
       <div>
           {/* Fetch Button */}
           <div style={{ textAlign: "center", marginTop: 20 }}>
               <button
                   onClick={handleFetchLogs}
                   style={{
                       padding: "10px 20px",
```

```
                        borderRadius: 6,
                        backgroundColor: "#1db954",
                        color: "white",
                        border: "none",
                        cursor: "pointer"
                    }}
                    disabled={status === "loading"}
                >
                    Fetch Logs
                </button>
            </div>

            {/* High Carbon */}
            <div style={{
                marginLeft: 'auto',
                backgroundColor: "#323232ff",
                marginRight: "auto",
                maxWidth: "30%",
                padding: 20,
                marginBottom: 10,
                marginTop: 10,
                borderRadius: 10,
                border: "2px solid red"
            }}>
                <h2>High Carbon Emission</h2>
                {logs
                    .filter(ele => ele.carbon >= 4)
                    .map((ele, index) => (
                        <h3 key={index} style={{ color: "red"
}}>
                            {ele.activity} {ele.carbon}
                        </h3>
                    ))}
            </div>

            {/* Low Carbon */}
            <div style={{
                marginLeft: 'auto',
                backgroundColor: "#323232ff",
                marginRight: "auto",
                maxWidth: "30%",
                padding: 20,
```

```
                borderRadius: 10,
                border: "2px solid green"
            }}>
                <h2>Low Carbon Emission</h2>
                {logs
                    .filter(ele => ele.carbon < 4)
                    .map((ele, index) => (
                        <h3 key={index} style={{ color: "green"
}}>
                            {ele.activity} {ele.carbon}
                        </h3>
                    ))}
            </div>
        </div>
    );
};

export default Logs;
```

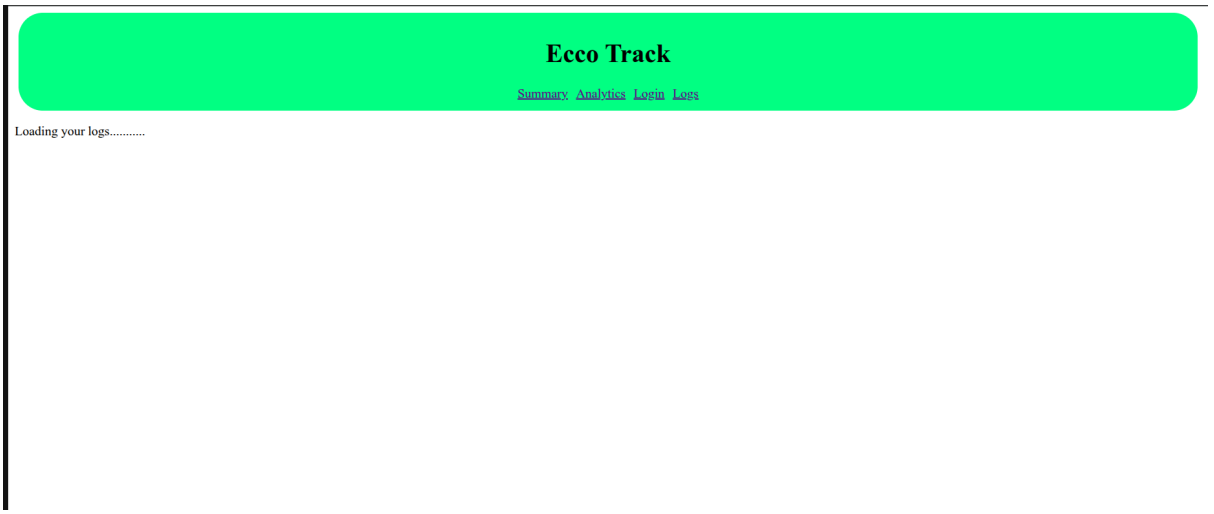
d. Logs.jsx

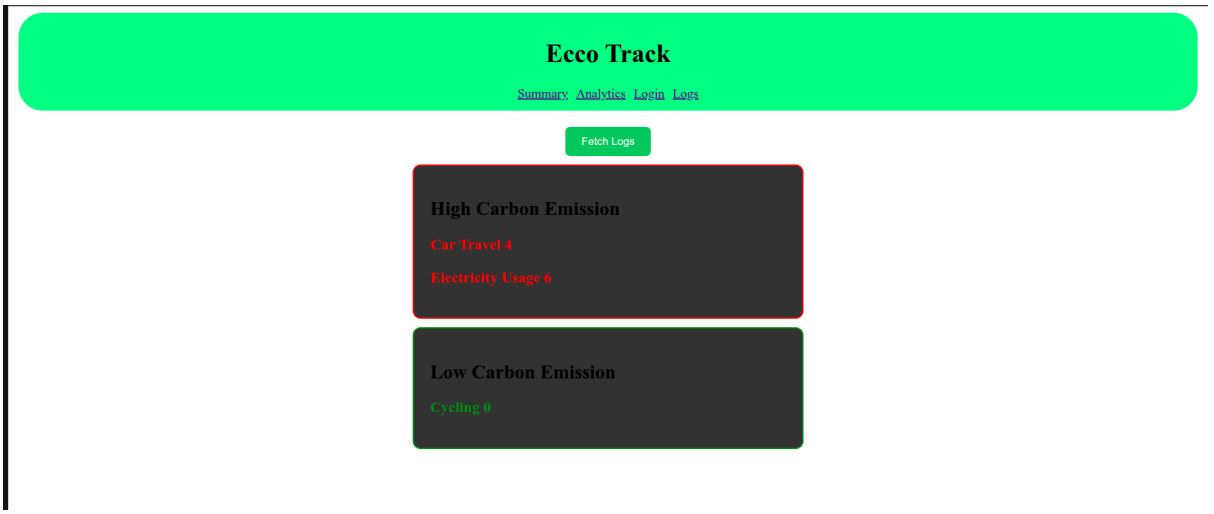
```
import { useEffect, useMemo, useCallback } from 'react';
import { useDispatch, useSelector } from 'react-redux';
import { fetchLogs } from '../store/logsSlice';

const Logs = () => {
    const dispatch = useDispatch();
    const { data = [], status, error } = useSelector((state) =>
state.logs);

    const handleFetchLogs = useCallback(() => {
        dispatch(fetchLogs());
    }, [dispatch]);

    useEffect(() => {
        if (status === 'idle') {
            dispatch(fetchLogs());
        }
    }, [status, dispatch]);

    const totalCarbon = useMemo(() => {
```

```
        return (data || []).reduce((acc, log) => acc +
(log.carbon || 0), 0);
    }, [data]);

    if (status === 'loading') {
        return <p>Loading your logs..........</p>;
    }

    if (status === 'failed') {
        return <p>Error: {error}</p>;
    }

    const logs = data || [];

    return (
        <div>
            <div style={{ textAlign: 'center', marginTop: 20 }}>
                <button
                    onClick={handleFetchLogs}
                    style={{
                        padding: '10px 20px',
                        borderRadius: 6,
                        backgroundColor: '#1db954',
                        color: 'white',
                        border: 'none',
                        cursor: 'pointer'
                    }}
                    disabled={status === 'loading'}
                >
                    Fetch Logs
                </button>
            </div>

            <div style={{ textAlign: 'center', marginTop: 8 }}>
                <h3>Total Carbon: {totalCarbon} kg CO2</h3>
            </div>

            <div style={{
                marginLeft: 'auto',
                backgroundColor: '#323232ff',
                marginRight: 'auto',
                maxWidth: '30%',
```

```
                padding: 20,
                marginBottom: 10,
                marginTop: 10,
                borderRadius: 10,
                border: '2px solid red'
            }}>
                <h2>High Carbon Emission</h2>
                {logs
                    .filter(ele => ele.carbon >= 4)
                    .map((ele, index) => (
                        <h3 key={index} style={{ color: 'red'
}}>
                            {ele.activity} {ele.carbon}
                        </h3>
                    ))}
            </div>

            <div style={{
                marginLeft: 'auto',
                backgroundColor: '#323232ff',
                marginRight: 'auto',
                maxWidth: '30%',
                padding: 20,
                borderRadius: 10,
                border: '2px solid green'
            }}>
                <h2>Low Carbon Emission</h2>
                {logs
                    .filter(ele => ele.carbon < 4)
                    .map((ele, index) => (
                        <h3 key={index} style={{ color: 'green'
}}>
                            {ele.activity} {ele.carbon}
                        </h3>
                    ))}
            </div>
        </div>
    );
};

export default Logs;
```

4. Output:


Ecco Track
Summary Analytics Login Logs
Loading your logs.....


Ecco Track
Summary Analytics Login Logs
Fetch Logs
High Carbon Emission
Car Travel 4
Electrielty Usage 6
Low Carbon Emission
Cycling 0

Ecco Track

Summary Analytics Login Logs

Login

Login Now

Ecco Track

Summary Analytics Login Logs

Dashboard_layout

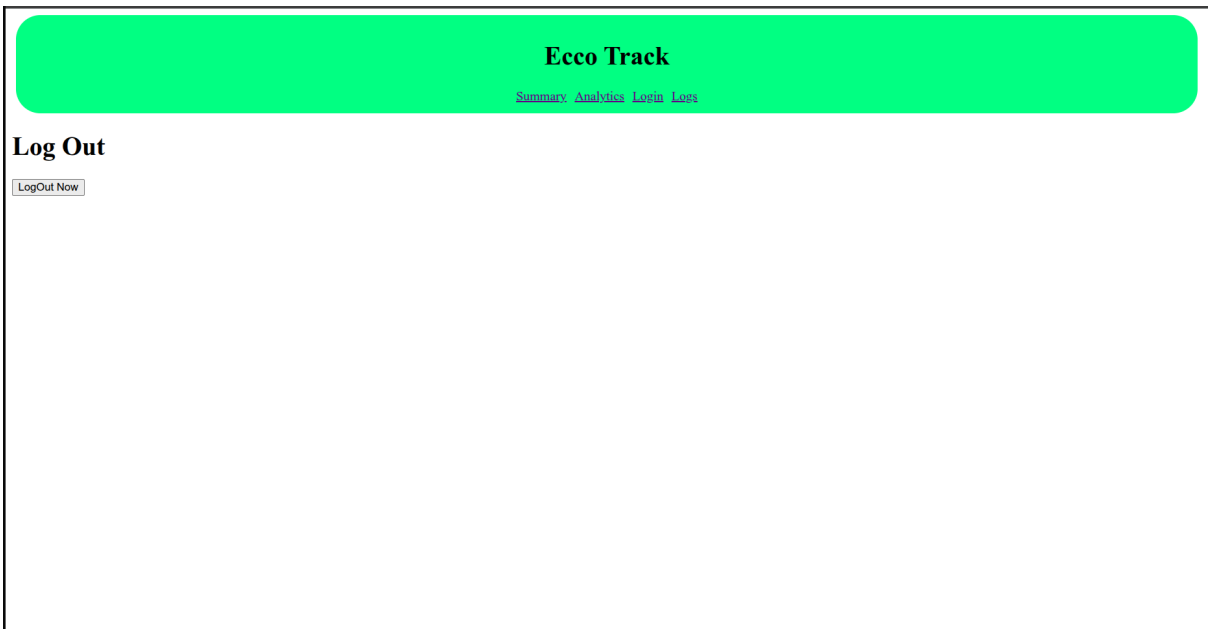
DashboardSummary

Ecco Track

Summary Analytics Login Logs

Dashboard_layout

DashboardAnalytics



5. Learning Outcome:

a. I learnt how to make redux react.
b. How the react redux and redux slices actually work.
c. I learnt how to use the outlet tags along with nested routes.